INCH-POUND

MIL-B-70838 (AR)
AMENDMENT 1
15 August 1991

MILITARY SPECIFICATION

BILLET, DEPLETED URANIUM

This amendment forms a part of Military Specification MIL-B-70838 (AR), dated 19 May 1989, and is approved for use by the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

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3.3.2 Records. Delete "UDI-R-22706A" and substitute "DI-T-2072".

3.4 Chemistry.

Delete: "Uranium^{235} (U^{235}) 0.25 weight percent maximum
Uranium (U) 99.95 weight percent minimum"

and substitute:

"Uranium (U) 99.94 weight percent minimum".

3.6 Matrix inclusions. Delete:

"3.6 Matrix inclusions. Billets shall be free from matrix inclusions larger than 0.0008 inch (0.080 inch at 100X) with a concentration not to exceed 11 inclusions per 0.02-inch square (2-inch square at 100X)."

and substitute:

"3.6 Matrix inclusions. Billets shall be free from matrix inclusions larger than 0.0008 inch (0.080 inch at 100X) with a concentration not to exceed 176 inclusions per 0.02-inch square (2-inch square at 100X)."

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3.11 Plate producibility.

Delete: "Hardness Rockwell A 62.0 - 68.0"

and substitute:

"Hardness Rockwell A 60.0 - 68.0".

Delete: "Transverse Percent Elongation 20 percent, minimum"

and substitute:

"Transverse Percent Elongation 12 percent, minimum".

Delete: "Longitudinal Yield Strength 68,000 psi, minimum"

and substitute:

"Longitudinal Yield Strength 57,000 psi, minimum".

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4.5.1.1 Hydrogen content. Delete "UDI-R-22706A/T" and substitute "DI-T-2072/T".

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4.5.1.2 Chemical analysis. Delete "UDI-R-22706A/T" and substitute "DI-T-2072/T".

4.5.1.4 Ultrasonic test. Delete "UDI-R-22706A/T" and substitute: "DI-T-2072/T".

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4.5.1.5.1.1 Transverse macrostructure. Delete "UDI-R-22706A/T" and substitute "DI-T-2072/T".

4.5.1.5.1.2 Longitudinal 1 macrostructure. Delete "UDI-R-22706A/T" and substitute "DI-T-2072/T".

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4.5.1.5.1.2 Longitudinal 2 macrostructure.

Delete: "4.5.1.5.1.2 Longitudinal 2 macrostructure."

and substitute:

"4.5.1.5.1.3 Longitudinal 2 macrostructure."

Delete "UDI-R-22706A/T" and substitute "DI-T-2072/T".

4.5.1.5.2 Microstructure. Delete "UDI-R-22706A/T" and substitute "DI-T-2072/T".

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4.5.1.6 Matrix inclusions. Delete:

"4.5.1.6 Matrix inclusions. A sample shall be metallographically prepared using standard laboratory procedures from the bottom of the top crop of each billet contained in the first article sample. These samples shall be viewed at 100X under bright field illumination and a photomicrograph taken of each sample. Photomicrographs shall be taken from the region of the sample containing the highest density of inclusions. The photomicrographs will be examined to determine compliance with 3.6. A copy of each photomicrograph from the first article sample shall be submitted to the Government in accordance with UDI-R-22706A."

and substitute:

"4.5.1.6 Matrix inclusions. A sample shall be metallographically prepared using standard laboratory procedures from the bottom of the top crop of each billet contained in the first article sample. These samples shall be viewed at 100X under bright field illumination and a photomicrograph taken of each sample. Photomicrographs shall be taken from the region of the sample containing the highest density of inclusions. Using a 2-inch square template divided into 0.5-inch squares, the region of the photomicrograph displaying the highest density of inclusions shall be identified and the number of inclusions counted by tallying the number of inclusions contained within each one-half inch square. Failure of any sample to comply with the requirements of 3.6 shall be cause for rejection of the first article. A copy of each photomicrograph from the first article sample shall be submitted to the Government in accordance with DI-T-2072."

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4.5.1.7.1 Surface examination. Delete:

"4.5.1.7.1 Surface examination. The plates rolled from the first article sample shall be visually examined for defects cited in 3.7. If more than five percent (5%) of the plates rolled from the first article sample exhibit rejectable defects (see 3.7) the first article shall be rejected."

and substitute:

"4.5.1.7.1 Surface examination. The plates produced from the first article sample shall be visually examined for the defects cited in 3.8. If more than five percent (5%) of the plates produced from the first article sample exhibit rejectable defects (see 3.8) the first article shall be rejected."

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4.5.1.7.2.3.2 Acceptance. Delete:

"a. All valid (non-disregarded) test results for each analysis (tensile strength, yield strength and percent elongation) in each alignment (longitudinal and transverse) for each plate shall meet the requirements of 3.11;"

and substitute:

"a. No more than two (2) valid (non-disregarded) test results from each plate shall fail to meet the requirements of

separately. For each lot, the sample pigs selected for tests shall be broken so as to present a clean fracture over the cross section of the pigs. One-half of each pig shall have drillings taken from three points triangularly spaced on the clean part of the fracture. One of the three drilling points shall be located half way from the center of the cross section to the bottom of the pig; another point shall be located half way between the center of the cross section and a top corner of the pig. The other point shall be located half way between the center of the cross section and other top corner of the pig. An equal proportion of drillings shall be taken from each hole in each sample pig of the lot. These drillings shall be combined, thoroughly mixed, and an analysis shall be made from the mixed drillings. All drillings shall be fine, clean, dry, and free from foreign substances. Drillings shall be taken at slow speed and shall not be oxidized.

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and substitute:

"Using a 2-inch square template divided into 0.5-inch squares, the region of the photomicrograph displaying the highest density of inclusions shall be identified and the number of inclusions counted by tallying the number of inclusions contained within each one-half inch square. Failure of any sample to comply with the requirements of 3.6 shall be cause for rejection of all billets represented by the sample. A copy of each photo- micrograph from the first article sample shall be submitted to the Government in accordance with DI-T-2072."

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TABLE II. Data Requirements. Delete "UDI-R-22706A" (three places) and substitute "DI-T-2072" (three places).

Custodian:
Army - AR

Preparing activity:
Army - AR

(Project 9650-A061)

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